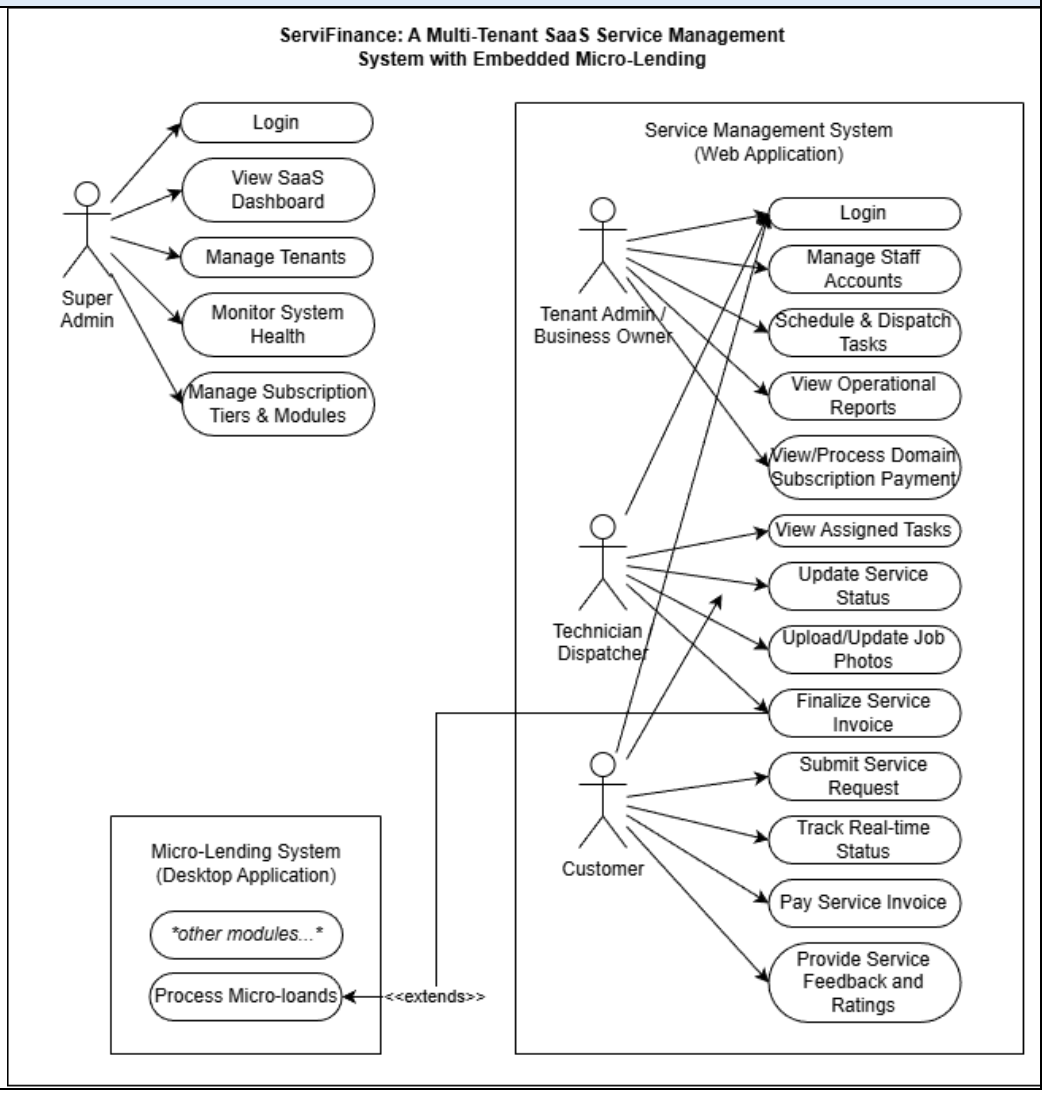
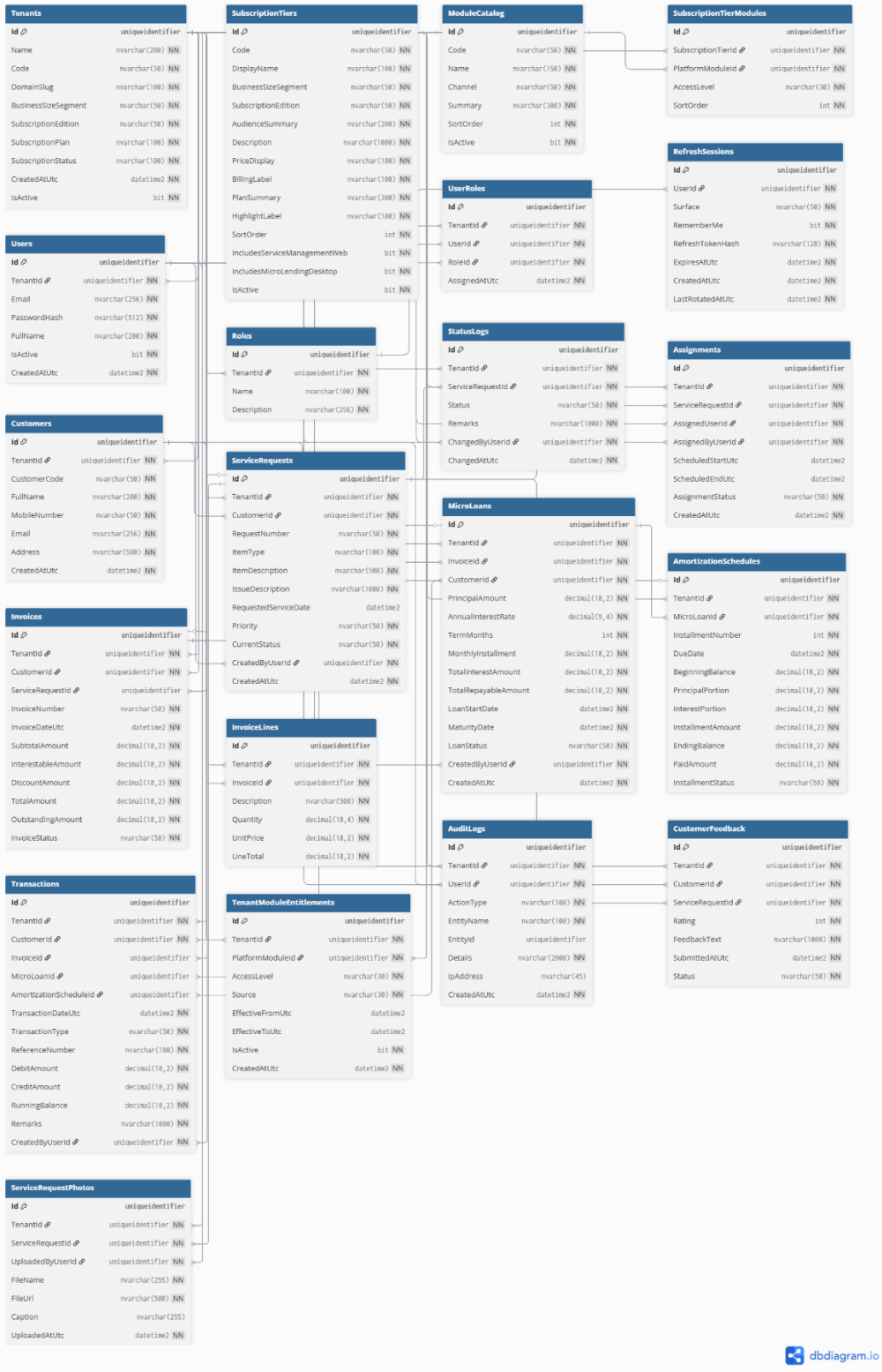
NAME	**CHRISTOPHER LLANDER L. VILLACINO**
PROJECT TITLE	**ServiFinance: A Multi-Tenant SaaS Service Management System with Embedded Micro-Lending**
SUBJECT: **CODE:** **TIME:**	**IT15/L Integrative Programming and Technologies** **8466** **1:30 – 3:30 PM**
TOPIC (Type of Business Process)	**#33 Service Management System**
Products/Services	**SaaS platform for Service-Oriented MSMEs**

1st Deliverables
AGILE MODEL: REQUIREMENTS AND PLANNING

Use Case Diagram (Updated)



Entity Relational Diagram	
Data Dictionary	**ServiFinance: A Multi-Tenant SaaS Service Management System with Embedded Micro-Lending**

Level 1 - Super Admin

Tenants table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the tenant record
Name	nvarchar	200	Tenant or business display name
Code	nvarchar	50	Unique business code used by the platform
DomainSlug	nvarchar	100	Unique route slug used for tenant URLs
BusinessSizeSegment	nvarchar	50	MSME size segment such as Micro, Small, or Medium
SubscriptionEdition	nvarchar	50	Product edition such as Standard or Premium
SubscriptionPlan	nvarchar	100	Commercial plan label for the tenant
SubscriptionStatus	nvarchar	100	Current subscription lifecycle state
CreatedAtUtc	datetime2	-	UTC date and time when the tenant was created
IsActive	bit	1	Active or inactive status of the tenant

SubscriptionTiers table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the subscription tier
Code	nvarchar	50	Unique catalog code of the tier
DisplayName	nvarchar	100	User-facing name of the tier
BusinessSizeSegment	nvarchar	50	MSME segment covered by the tier
SubscriptionEdition	nvarchar	50	Edition covered by the tier
AudienceSummary	nvarchar	200	Short target audience summary
Description	nvarchar	1000	Full description of the tier
PriceDisplay	nvarchar	100	Price label shown in the UI
BillingLabel	nvarchar	100	Billing period or billing note
PlanSummary	nvarchar	300	Short summary of plan inclusions
HighlightLabel	nvarchar	100	Badge or marketing highlight label
SortOrder	int	10	Display order in the catalog
IncludesServiceManagementWeb	bit	1	Indicates if the web service module is included
IncludesMicroLendingDesktop	bit	1	Indicates if the desktop lending module is included
IsActive	bit	1	Active or inactive catalog state

ModuleCatalog table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the module catalog row
Code	nvarchar	50	Unique module code such as W1 or D1
Name	nvarchar	150	Module name shown to operators
Channel	nvarchar	50	Delivery channel such as Web or Desktop
Summary	nvarchar	300	Short purpose statement of the module
SortOrder	int	10	Display order in the module catalog
IsActive	bit	1	Active or inactive catalog state

SubscriptionTierModules table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the tier-module mapping
SubscriptionTierId	uniqueidentifier	36	Foreign key to `SubscriptionTiers.Id`
PlatformModuleId	uniqueidentifier	36	Foreign key to `ModuleCatalog.Id`
AccessLevel	nvarchar	30	Access rule such as Included, Limited, or Not Included
SortOrder	int	10	Display order within the tier mapping

Users table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the user account
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`; platform superadmins use the reserved platform tenant
Email	nvarchar	256	Login email address of the user
PasswordHash	nvarchar	512	Hashed password value
FullName	nvarchar	200	Full name of the user
IsActive	bit	1	Active or inactive status of the account
CreatedAtUtc	datetime2	-	UTC date and time when the account was created

Roles table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the role
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`

Name	nvarchar	100	Role name such as SuperAdmin, Administrator, or Staff
Description	nvarchar	256	Short description of the role purpose

UserRoles table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the role assignment
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
UserId	uniqueidentifier	36	Foreign key to `Users.Id`
RoleId	uniqueidentifier	36	Foreign key to `Roles.Id`
AssignedAtUtc	datetime2	-	UTC date and time when the role was assigned

RefreshSessions table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the refresh session
UserId	uniqueidentifier	36	Foreign key to `Users.Id`
Surface	nvarchar	50	Client surface such as web or desktop
RememberMe	bit	1	Indicates whether a persistent session was requested
RefreshTokenHash	nvarchar	128	Unique hash of the refresh token
ExpiresAtUtc	datetime2	-	UTC date and time when the refresh session expires
CreatedAtUtc	datetime2	-	UTC date and time when the session was issued
LastRotatedAtUtc	datetime2	-	UTC date and time when the refresh token was last rotated

TenantModuleEntitlements table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the tenant-specific module entitlement
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
PlatformModuleId	uniqueidentifier	36	Foreign key to `ModuleCatalog.Id`
AccessLevel	nvarchar	30	Effective access level for the tenant
Source	nvarchar	30	Entitlement source such as tier-default, override, or promo
EffectiveFromUtc	datetime2	-	Optional start date of the entitlement
EffectiveToUtc	datetime2	-	Optional end date of the entitlement
IsActive	bit	1	Indicates whether the entitlement is currently active
CreatedAtUtc	datetime2	-	UTC date and time when the entitlement was recorded

AuditLogs table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the audit log record
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
UserId	uniqueidentifier	36	Foreign key to `Users.Id` for the actor
ActionType	nvarchar	100	Action category such as Login, ProvisionTenant, or UpdateRole
EntityName	nvarchar	100	Name of the affected table or business entity
EntityId	uniqueidentifier	36	Identifier of the affected record when applicable
Details	nvarchar	2000	Full audit details or serialized change summary
IpAddress	nvarchar	45	Source IP address if captured
CreatedAtUtc	datetime2	-	UTC date and time when the action happened

Level 2 - Tenant Administrator / Owner
Users table (*refer to Level 1 – Super Admin)
Roles table (*refer to Level 1 – Super Admin)
UserRoles table (*refer to Level 1 – Super Admin)

Customers table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the customer
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
CustomerCode	nvarchar	50	Unique customer code within the tenant
FullName	nvarchar	200	Full name of the customer
MobileNumber	nvarchar	50	Mobile or contact number
Email	nvarchar	256	Customer email address
Address	nvarchar	500	Customer address
CreatedAtUtc	datetime2	-	UTC date and time when the customer was registered

ServiceRequests table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the service request
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
CustomerId	uniqueidentifier	36	Foreign key to `Customers.Id`
RequestNumber	nvarchar	50	Unique service request number within the tenant
ItemType	nvarchar	100	General item type, such as oven or laptop

ItemDescription	nvarchar	500	Detailed description of the service item
IssueDescription	nvarchar	1000	Customer-reported issue or complaint
RequestedServiceDate	datetime2	-	Requested service schedule, if provided
Priority	nvarchar	50	Priority level of the request
CurrentStatus	nvarchar	50	Current workflow status of the request
CreatedByUserId	uniqueidentifier	36	Foreign key to `Users.Id` for the creator
CreatedAtUtc	datetime2	-	UTC date and time when the request was created

StatusLogs table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the status log
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
ServiceRequestId	uniqueidentifier	36	Foreign key to `ServiceRequests.Id`
Status	nvarchar	50	Status value recorded for the request
Remarks	nvarchar	1000	Remarks or explanation of the status change
ChangedByUserId	uniqueidentifier	36	Foreign key to `Users.Id` for the operator who changed the status
ChangedAtUtc	datetime2	-	UTC date and time when the status changed

Assignments table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the assignment
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
ServiceRequestId	uniqueidentifier	36	Foreign key to `ServiceRequests.Id`
AssignedUserId	uniqueidentifier	36	Foreign key to `Users.Id` for the assigned staff member
AssignedByUserId	uniqueidentifier	36	Foreign key to `Users.Id` for the assigning manager
ScheduledStartUtc	datetime2	-	Planned start date and time
ScheduledEndUtc	datetime2	-	Planned end date and time
AssignmentStatus	nvarchar	50	Current status of the assignment
CreatedAtUtc	datetime2	-	UTC date and time when the assignment was created

Invoices table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the invoice

TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
CustomerId	uniqueidentifier	36	Foreign key to `Customers.Id`
ServiceRequestId	uniqueidentifier	36	Optional foreign key to `ServiceRequests.Id`
InvoiceNumber	nvarchar	50	Unique invoice number within the tenant
InvoiceDateUtc	datetime2	-	UTC date and time of invoice issuance
SubtotalAmount	decimal	18,2	Sum of invoice line amounts before discount and loan handling
InterestableAmount	decimal	18,2	Amount eligible for financing or interest
DiscountAmount	decimal	18,2	Discount amount applied to the invoice
TotalAmount	decimal	18,2	Final total amount due
OutstandingAmount	decimal	18,2	Remaining unpaid amount
InvoiceStatus	nvarchar	50	Status of the invoice, such as Open or Paid

InvoiceLines table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the invoice line
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
InvoiceId	uniqueidentifier	36	Foreign key to `Invoices.Id`
Description	nvarchar	500	Description of the billed item or service
Quantity	decimal	18,4	Quantity billed on the line
UnitPrice	decimal	18,2	Unit price of the line item
LineTotal	decimal	18,2	Extended amount of the line item

MicroLoans table (Desktop side)

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the micro-loan
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
InvoiceId	uniqueidentifier	36	Foreign key to `Invoices.Id`; one invoice can produce at most one loan
CustomerId	uniqueidentifier	36	Foreign key to `Customers.Id`
PrincipalAmount	decimal	18,2	Loan principal amount
AnnualInterestRate	decimal	9,4	Annual interest rate used for computation
TermMonths	int	10	Loan term in months
MonthlyInstallment	decimal	18,2	Fixed installment amount per month
TotalInterestAmount	decimal	18,2	Total interest for the whole loan
TotalRepayableAmount	decimal	18,2	Total amount to be repaid

LoanStartDate	datetime2	-	Start date of the loan
MaturityDate	datetime2	-	Final due date of the loan
LoanStatus	nvarchar	50	Current status of the loan
CreatedByUserId	uniqueidentifier	36	Foreign key to `Users.Id` for the creating operator
CreatedAtUtc	datetime2	-	UTC date and time when the loan was created

AmortizationSchedules table (Desktop side)

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the amortization row
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
MicroLoanId	uniqueidentifier	36	Foreign key to `MicroLoans.Id`
InstallmentNumber	int	10	Sequence number of the installment
DueDate	datetime2	-	Due date of the installment
BeginningBalance	decimal	18,2	Loan balance before the installment
PrincipalPortion	decimal	18,2	Principal amount included in the installment
InterestPortion	decimal	18,2	Interest amount included in the installment
InstallmentAmount	decimal	18,2	Total installment amount due
EndingBalance	decimal	18,2	Balance after the installment
PaidAmount	decimal	18,2	Amount already paid against the installment
InstallmentStatus	nvarchar	50	Status of the installment, such as Unpaid, Partial, or Paid

Transactions table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the ledger transaction
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
CustomerId	uniqueidentifier	36	Foreign key to `Customers.Id`
InvoiceId	uniqueidentifier	36	Optional foreign key to `Invoices.Id`
MicroLoanId	uniqueidentifier	36	Optional foreign key to `MicroLoans.Id`
AmortizationScheduleId	uniqueidentifier	36	Optional foreign key to `AmortizationSchedules.Id`
TransactionDateUtc	datetime2	-	UTC date and time of the ledger event
TransactionType	nvarchar	50	Type of ledger event, such as Invoice, Loan, or Payment

ReferenceNumber	nvarchar	100	Human-readable transaction reference
DebitAmount	decimal	18,2	Debit amount recorded in the ledger
CreditAmount	decimal	18,2	Credit amount recorded in the ledger
RunningBalance	decimal	18,2	Balance after the transaction posting
Remarks	nvarchar	1000	Transaction remarks or explanation
CreatedByUserId	uniqueidentifier	36	Foreign key to `Users.Id` for the creating operator

CustomerFeedback table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the customer feedback record
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
CustomerId	uniqueidentifier	36	Foreign key to `Customers.Id`
ServiceRequestId	uniqueidentifier	36	Foreign key to `ServiceRequests.Id`
Rating	int	10	Numeric service rating submitted by the customer
FeedbackText	nvarchar	1000	Comment or feedback text
SubmittedAtUtc	datetime2	-	UTC date and time when feedback was submitted
Status	nvarchar	50	Review status of the feedback

ServiceRequestPhotos table

Field Names	Datatype	Length	Description
Id	uniqueidentifier	36	Primary key of the service request photo record
TenantId	uniqueidentifier	36	Foreign key to `Tenants.Id`
ServiceRequestId	uniqueidentifier	36	Foreign key to `ServiceRequests.Id`
UploadedByUserId	uniqueidentifier	36	Foreign key to `Users.Id` for the uploader
FileName	nvarchar	255	Original or stored file name
FileUrl	nvarchar	500	File path or blob URL of the uploaded image
Caption	nvarchar	255	Optional caption or note for the image
UploadedAtUtc	datetime2	-	UTC date and time when the image was uploaded

Level 3 - Service Staff / Technician
Tables that are identical to Level 2 - Tenant Administrator / Owner:

- **Customers table**
- **ServiceRequests table**

	• **StatusLogs table** • **Assignments table** • **ServiceRequestPhotos table** • **CustomerFeedback table** **Level 4 - Finance Staff / Loan Officer / Cashier (Desktop side)** **Tables that are identical to Level 2 - Tenant Administrator / Owner:** • **Customers table** • **Transactions table** • **AuditLogs table** • **Invoices table** • **InvoiceLines table** • **MicroLoans table** • **AmortizationSchedules table** **Level 5 - Customer** **Tables that are identical to Level 2 - Tenant Administrator / Owner:** • **Customers table** • **ServiceRequests table** • **Invoices table** • **CustomerFeedback table**
Date Submitted:	
Teacher's Feedback	_____ _____ _____ _____ _____ _____ _____ _____ Student's Signature (after feedbacking) Teacher's Signature